

Final Project

2023-12-05

Introduction

The definition of low birth weight can be found by one given by the institution known as the World Health Organization (W.H.O.). This definition considers any child with a birth weight of 2,500 grams or less (regardless of the gestational age of the child) to be a child of low birth weight (LBW). This is an important health condition to note for neonatal and other health care professionals as, whether or not a child is of low birth weight is a highly valuable public health indicator for the health of the mother, nutrition, healthcare, delivery, and poverty. Low birth weight newborns are at a higher overall risk of death and illness shortly after birth, and even non-communicable diseases later in life. Specifically, infants of low birth weight are 20 times more likely to develop complications and die in comparison to non-low birth weight infants. Newborns of low birth weight have a much higher risk of cognitive deficits, motor delays, cerebral palsy, and other psychological and behavioral problems. Although the exact causal factors at play are not completely known, the condition does statistically seem to have global prevalence. Low birth weight is currently estimated at 15 to 20 % of all births worldwide. As this condition is highly correlated to many ensuing health problems which bear an increased strain on the public health system, identifying ways to reduce the probability of children being born at low birth weight could not only be an effective method of reducing mortality rate and illness in newborns, but also may save the health care system and households money overall. As such we have investigated a publicly available dataset on low birth weight to gain a better understanding of several factors which can elucidate that which determines the probability of a child being born low birth weight or not.

Problem Statement & Background

The fact that infant mortality rates and birth defect rates are higher for low birth weight babies. We want to find how a woman's behavior during pregnancy (including diet, smoking habits, and receiving prenatal care) and how that can greatly alter the chances of carrying the baby to term, and, consequently, of delivering a baby of normal birth weight.

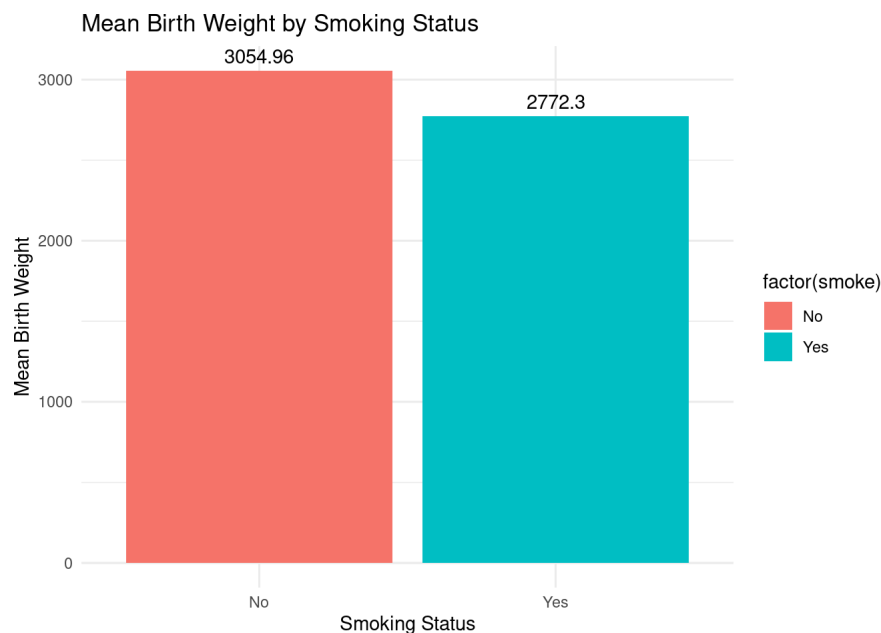
Sampling Method and collection instrument: A sample 189 births to women seen in the obstetrics clinic. Fifty-nine of these births were of low birth weight.

Data Analysis - Question 1

Smoke and Low Birth Contingency Table

	Birth Weight >= 2500	Birth Weight < 2500	Total
Smoke No	86	29	115
Smoke Yes	44	30	74
Total	130	59	189

Smoke vs Bwt Graph



Correlation Testing Smoke and Bwt

```
##
## Welch Two Sample t-test
##
## data: bwt by smoke
## t = 2.7192, df = 170.04, p-value = 0.007224
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  77.45839 487.86006
## sample estimates:
## mean in group No mean in group Yes
##      3054.957      2772.297
```

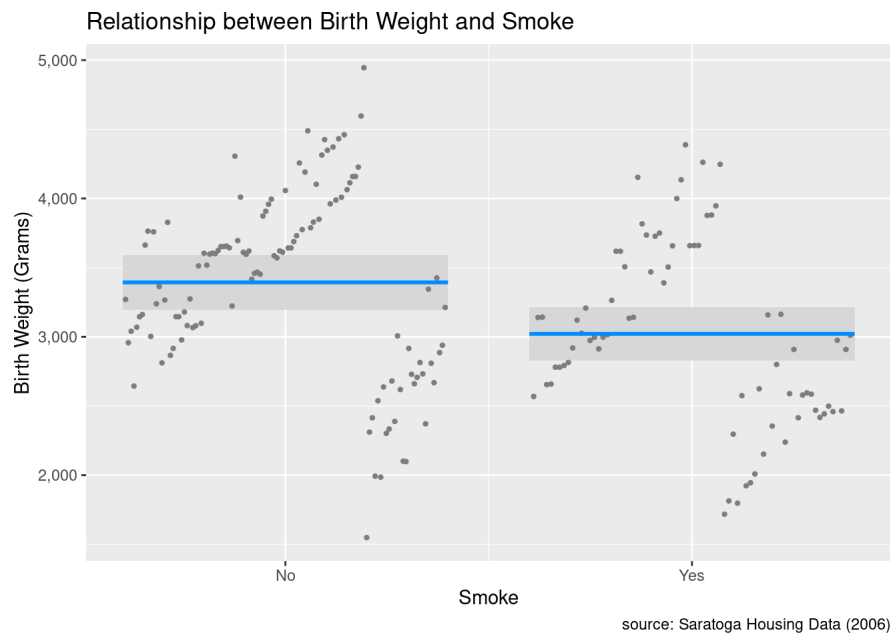
```
##
## Pearson's product-moment correlation
##
## data: bwt and numeric_category
## t = -2.6428, df = 187, p-value = 0.00892
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.32371283 -0.04832827
## sample estimates:
##      cor
## -0.1897497
```

```
## [1] -0.1897497
```

Regression Modeling Smoke and Bwt

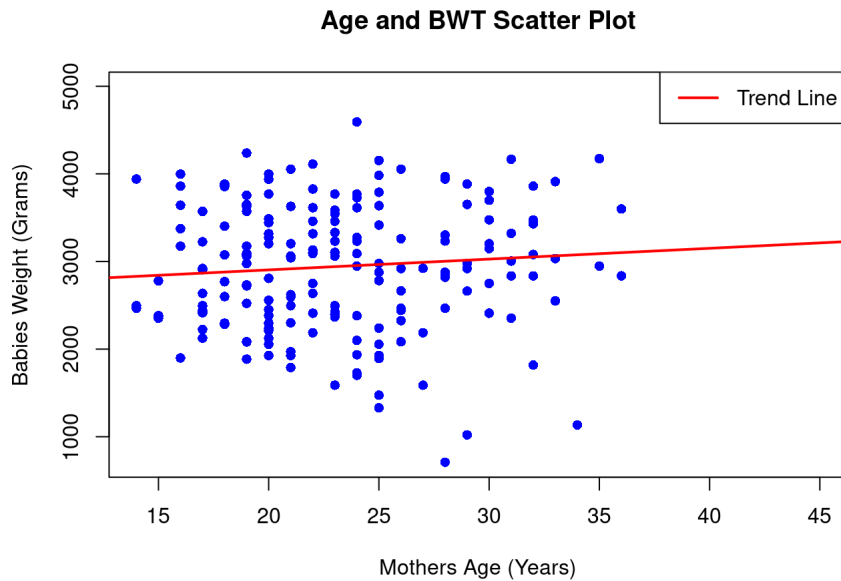
```
library(visreg)

visreg(data, "smoke", gg = TRUE) +
  scale_y_continuous(label = scales::comma) +
  labs(title = "Relationship between Birth Weight and Smoke",
       caption = "source: Saratoga Housing Data (2006)",
       y = "Birth Weight (Grams)",
       x = "Smoke")
```



Data Analysis - Question 2

Age and Birth Weight Scatter Plot



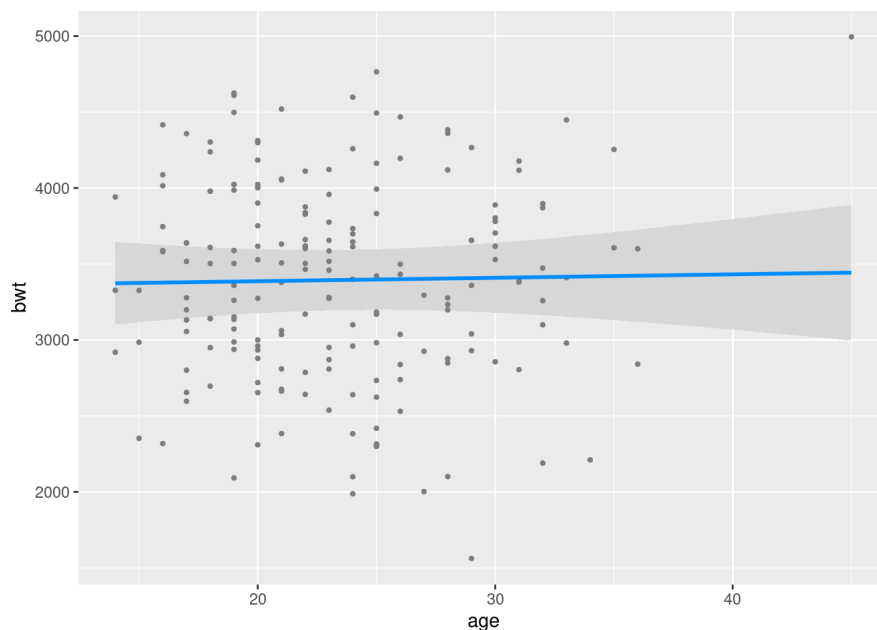
Correlation Testing Age and Bwt

```
##
## Two Sample t-test
##
## data: age and bwt
## t = -55.083, df = 376, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -3025.319 -2816.776
## sample estimates:
## mean of x mean of y
## 23.2381 2944.2857
```

```
##
## Pearson's product-moment correlation
##
## data: bwt and age
## t = 1.2289, df = 187, p-value = 0.2207
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.05391447 0.22930546
## sample estimates:
## cor
## 0.08950457
```

```
## [1] 0.08950457
```

Regression Modeling for Age and BWT



Data Analysis - Question 3 (Dori)

- Would mothers who have hypertension (ht), or high blood pressure, more likely give birth to babies with a low birth weight (low)?

To address this research question, we have created a contingency table between the two categorical variables hypertension(ht) and low birth weight (low) to visualize the frequencies of different combinations of the two variables. We also used a mosaic plot and a bar plot to see the graphical relations between the two variables. After the graphs, we performed a chi-squared test to compare the mean proportions between the two variables. See findings below.

CONTINGENCY TABLE

low	ht	Frequencies	Cell.Percentage	Row.Percentage	Column.Percentage
No	No	125	66.137566	70.62147	70.62147
No	Yes	5	2.645503	41.66667	29.37853
Yes	No	52	27.513227	29.37853	41.66667
Yes	Yes	7	3.703704	58.33333	58.33333

MOSAIC PLOT

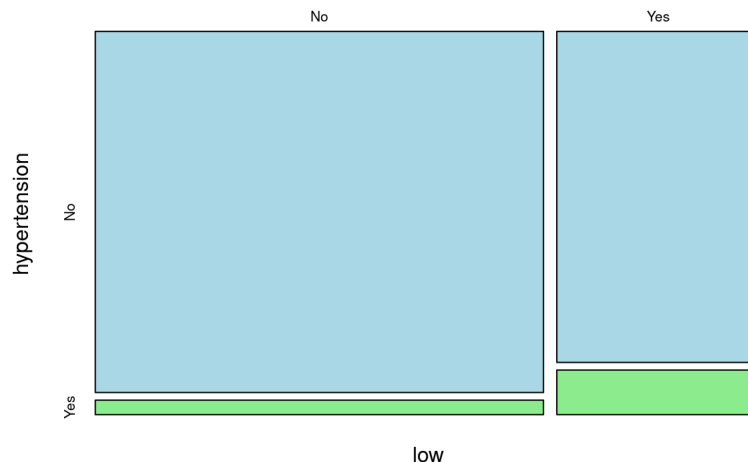
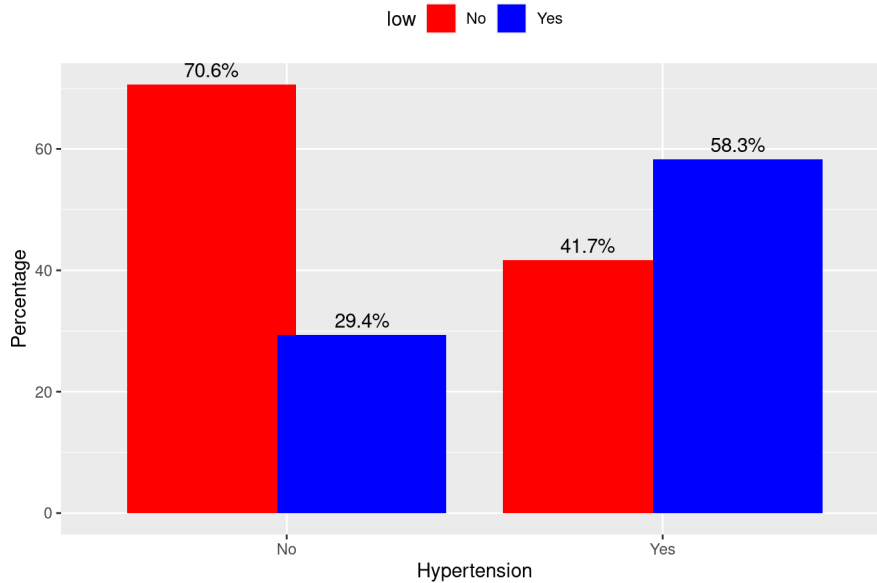


Figure 3.2: Mosaic Plot low vs ht

BARPLOT

Barplot by Hypertension and Low Birth Weight (Percentage by Row)



Comparing 2 Proportions

Null Hypothesis: $H_0 : \pi_1 = \pi_2$

- There is no significant difference in the two population proportions for the variables hypertension and low birth weight.

Alternate Hypothesis: $H_A : \pi_1 \neq \pi_2$

- There is a significant difference in the two population proportions for the variables hypertension and low birth weight.

Significant level: $\alpha = 0.05$

```
##
## 2-sample test for equality of proportions with continuity correction
##
## data: contingency_table
## X-squared = 3.1431, df = 1, p-value = 0.07625
## alternative hypothesis: two.sided
## 95 percent confidence interval:
## -0.04184104 0.62093709
## sample estimates:
## prop 1 prop 2
## 0.7062147 0.4166667
```

The p-value (0.07625) is greater than the significance level of 0.05. Based on this result, we do not have enough evidence to reject the null hypothesis that the two mean proportions are equal. The 95 percent confidence interval also indicates that the two proportions are equal, as the interval contains 0 (-0.04184104, 0.62093709).

Looking at the results from the bar graph and mosaic plot, having hypertension seems to increase the chances of giving birth to a baby with a low birth weight. As you can see, the chances of having a low birth weight baby is 29.4% if the mother does not have hypertension, but when the mother does the chances increase to 58.3%!

Finally, based on both the chi-squared test and graphical summaries, there is suggestive evidence that having hypertension may be associated as a risk factor in giving birth to a baby with low birth weight. While the statistical significance is not strong according to the chi-square test (might be due to our sample size), the bar plot and mosaic plot suggests a potential trend that hypertension is a likely risk factor associated with low birth weight.

Data Analysis - Question 4 (Dori)

- Does the mother's race affect the mother's weight at last menstrual cycle(lwt)?

To understand the relationship between a mother's race and her weight at last menstrual cycle (lwt), we performed the following analyses:

SUMMARY STATISTIC

Table 4

Race	n	mean	sd	min	max
Black	26	146.8077	39.63939	98	241
Other	67	120.0299	25.12150	80	250
White	96	132.0521	29.09381	90	235

From the numerical summary we can interpret that the mean last weight of mothers that are black and white are heavier than mothers of other races.

EXPLORATORY DATA

HISTOGRAM

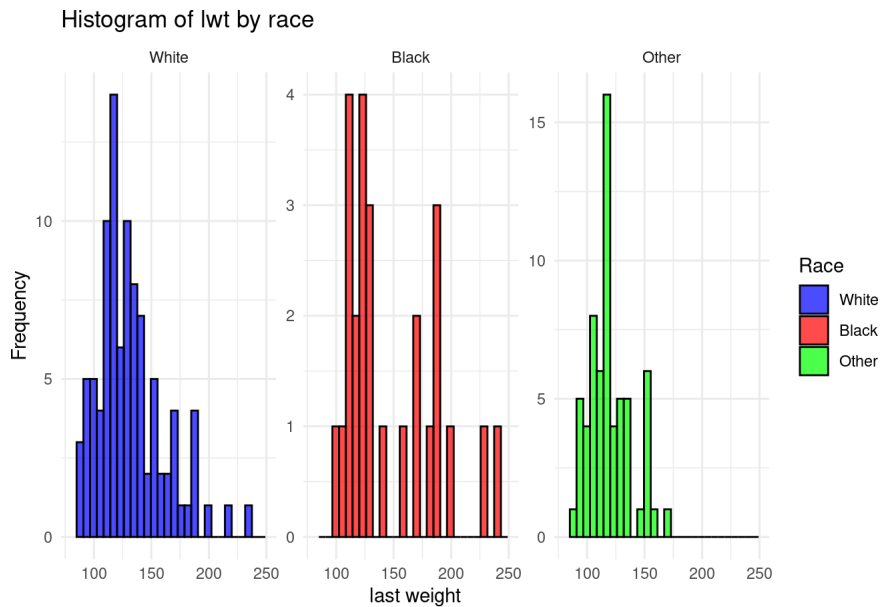


Figure 4.2

From Figure 4.2, we see histograms showing the distributions of last weights for mothers of races white, black, and other. Looking at the distributions, mothers who are black have a wider and more spread out distribution compared to mothers of other races. This shows how mothers of black races are typically heavier than mothers of other races. Mothers of white races are typically heavier than mothers of other races.

BOXPLOT

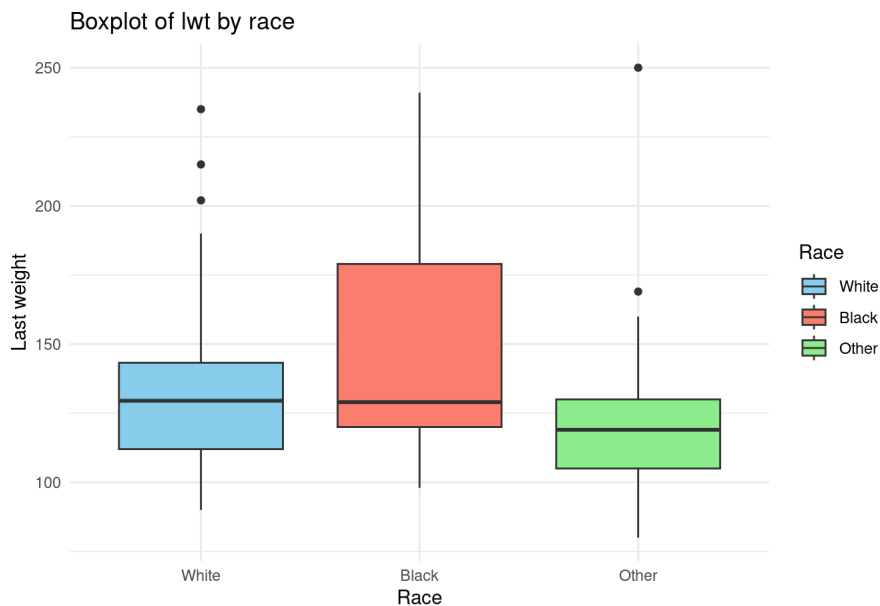


Figure 4.3

Looking at the box plot from Figure 4.3, we can interpret that the median last weights of mothers that are white and black are very similar. The last weight of mothers of other races are slightly lighter similar to what we see in our histograms from Figure 4.2.

Null Hypothesis (H₀): $H_0 : \mu_{white} = \mu_{black} = \mu_{other}$

- There is no difference in the average weight of mothers among different races.

Alternate Hypothesis (H_A): $H_A : \mu_{white} \neq (or) \mu_{black} \neq (or) \mu_{other}$

- The average weights among mothers are different for at least two of the races.

Significant level: $\alpha = 0.01$

ANOVA

Source	Sum_Sq	Df	Mean_Sq	F_value	Pr_gt_F
race	14403.17	2	7201.5826	8.301962	0.0003519
Residuals	161346.72	186	867.4555	NA	NA

After conducting an ANOVA test, we have statistical evidence that mothers of at least two races have significantly different average last weights before menstrual period as our p-value (0.000352) indicates this (very small p-value).

Underlying Assumptions

Null Hypothesis (H0):

- The population from which the sample is drawn follows a normal distribution.

Alternate Hypothesis (HA):

- The population from which the sample is drawn does not follow a normal distribution.

Significant level: $\alpha = 0.05$

NORMALITY

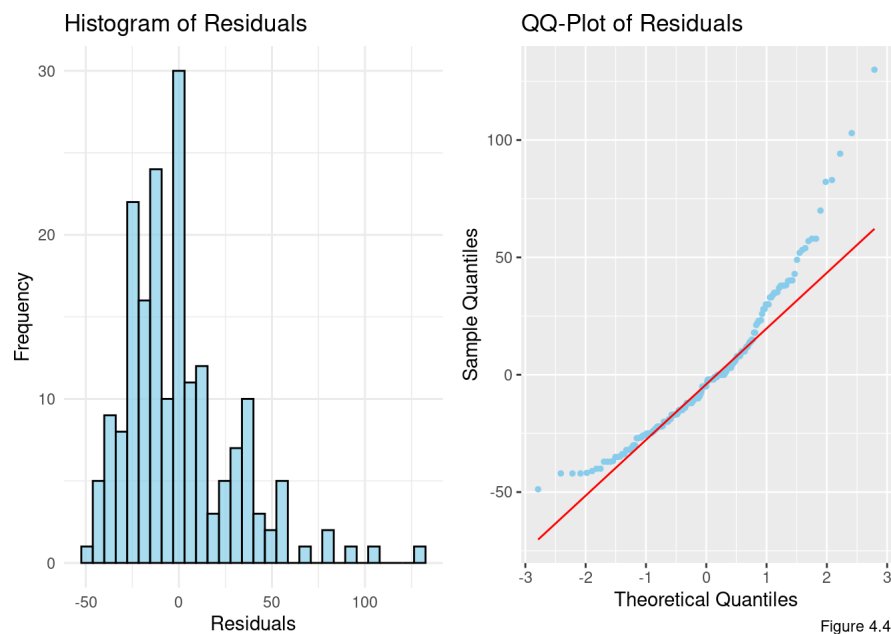


Figure 4.4 presents a histogram of residuals. From the histogram, it seems like the distribution is normally distributed as it seems like a normal bell-curved shape. Looking at the qq-plot results, we see that we have a slightly right skewed result, slightly deviating from the normal distribution.

SHAPIRO TEST RESULTS

	Statistic	P_Value
W	0.9162574	6.711319e-09

After performing a Shapiro-Wilk normality test, from the results we can assume that based on the small p-value (6.711319e-09), there is evidence to suggest that the residuals do not follow a normal distribution. We would reject the null hypothesis of normality or following a normal distribution.

Homogeneity of Variance

BARPLOT

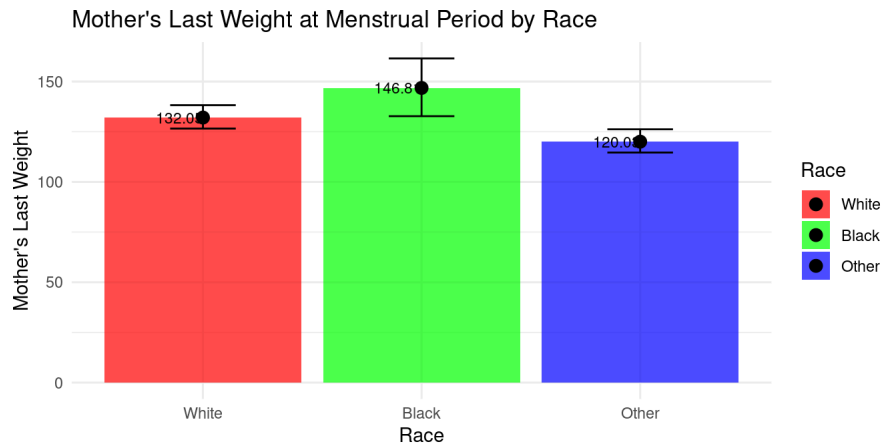


Figure 4.5

This bar plot (Figure 4.5) shows information about the means of races of mothers who are white, black, or other. The mean is marked by the dots. The lines on each bar represent a measure of variance around each mean as well.

LEVENE'S TEST

- F-Value: 4.2941
- P-Value: 0.01503

After performing a Levene's test, from the Levene's test, the p-value is 0.01503, which is less than 0.05 (our significance level is 0.05). This suggests that there is evidence to reject the null hypothesis of equal variances between groups.

BARTLETT'S TEST

- K-squared: 8.388
- P-Value: 0.01509

After performing a Bartlett test of homogeneity of variances, our p-value (0.01509) is less than 0.05, therefore we would reject the null hypothesis. This suggests that at least one group has a different variance.

INDEPENDENCE

DURBIN WATSON'S TEST

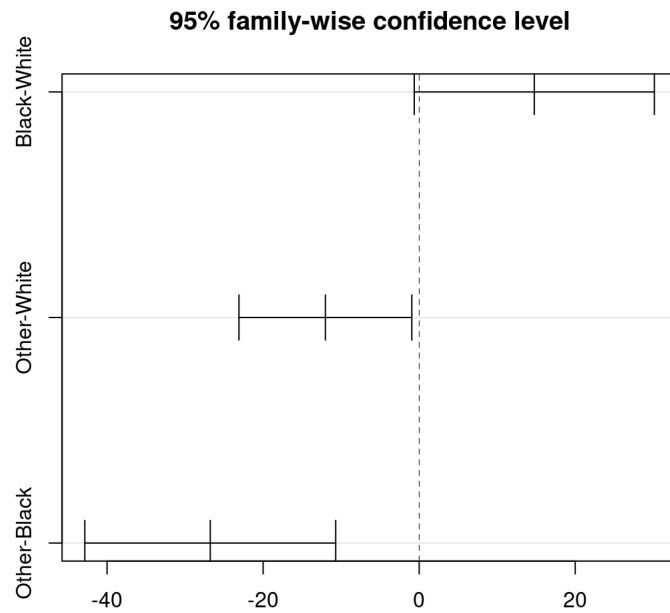
D-W Statistic: 2.01353 P-Value: 0.934

After performing a Durbin-Watson test for independence, we see from our p-value of 0.934 that it does not reject our null hypothesis since the p-value is very high.

COMPARISONS THROUGH TESTS

TUKEY TEST

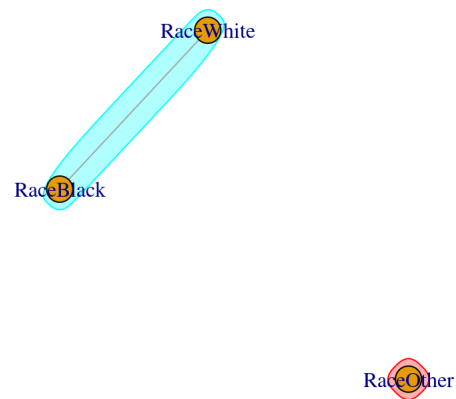
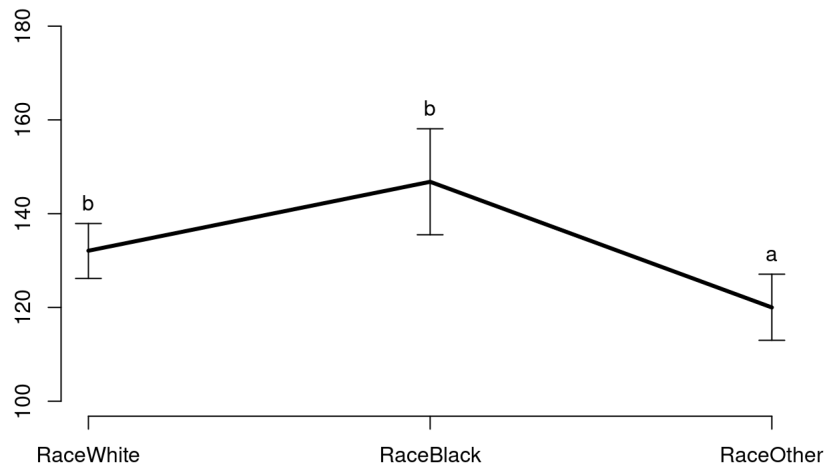
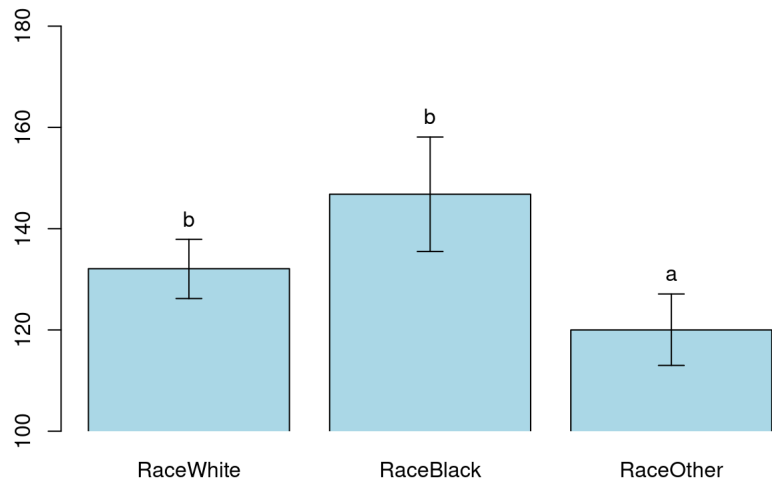
Comparison	diff	lwr	upr	p_adj
Black-White	14.75561	-0.6286544	30.1398723	0.0631935
Other-White	-12.02223	-23.0996843	-0.9447809	0.0298154
Other-Black	-26.77784	-42.8560065	-10.6996766	0.0003444



Looking at this graph, we see that the

Black-White has a confidence interval that contains the number 0. This indicates that there is not enough evidence to conclude a significant difference between the Black and White groups. For the other comparisons of Other-Black and Other-White, the interval does not contain 0, indicating that there is a significant difference.

POST HOC ANALYSIS



Looking at the results of the Tukey test, what we can conclude is that there is a significant difference between the races of other and white, and also a significant difference between the races of other and black. For mothers who were white and black, however, there is no strong evidence indicating for a significant difference as the p-value for the two is 0.0631935, which is greater than our significant level of 0.05. Therefore, we are unable to reject the null hypothesis for the races white and black, meaning that they are equal.

In the end, through the results of these analyses, we have evidence to suggest an association between the mother's race and her last weight at menstrual cycle. This is because our ANOVA test results indicate that there is a statistically significant difference in the average last weights among mothers of different races. Additionally, the post hoc analyses highlight specific differences between certain race groups. Therefore, we can suggest that race and last weight variables are associated with each other.

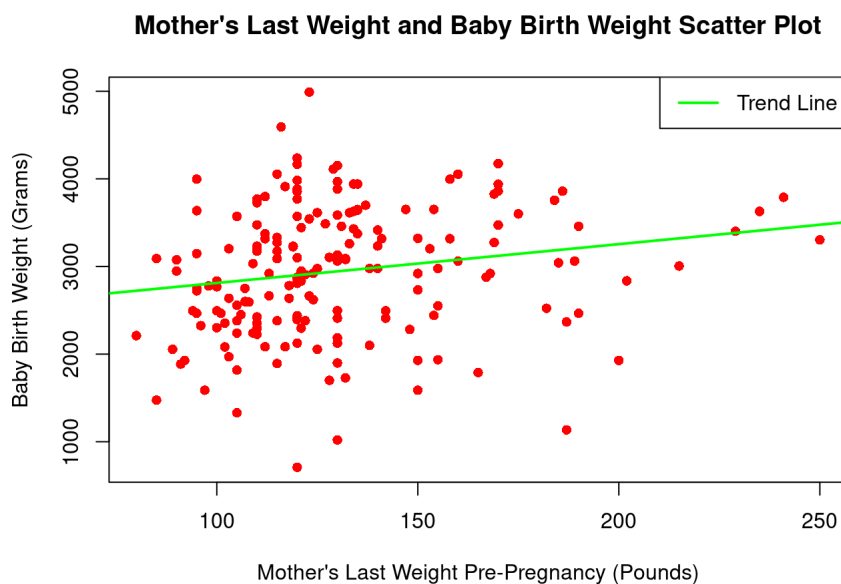
Data Analysis - Question 5 (Vinton)

Null Hypothesis (H₀): \$H_0\$: There is no statistically significant association between the mother's weight at last menstruation, and the birth weight of her child.

Alternate Hypothesis (H_A): \$H_A\$: There is a statistically significant association between the mother's weight at last menstruation, and the birth weight of her child.

As this is a question relating two quantitative variables, we should start to be able to get a good idea of the answer merely from plotting the variables against each other. Let's do this via a scatter plot.

Mother's last weight pre-pregnancy and Birth Weight Scatter Plot



no correlation between the two variables. Although a trendline can be constructed somewhere along it, its data visually seems to have no obvious relationship. Let's compute some some statistical values to be sure though.

Correlation Testing Age and Bwt

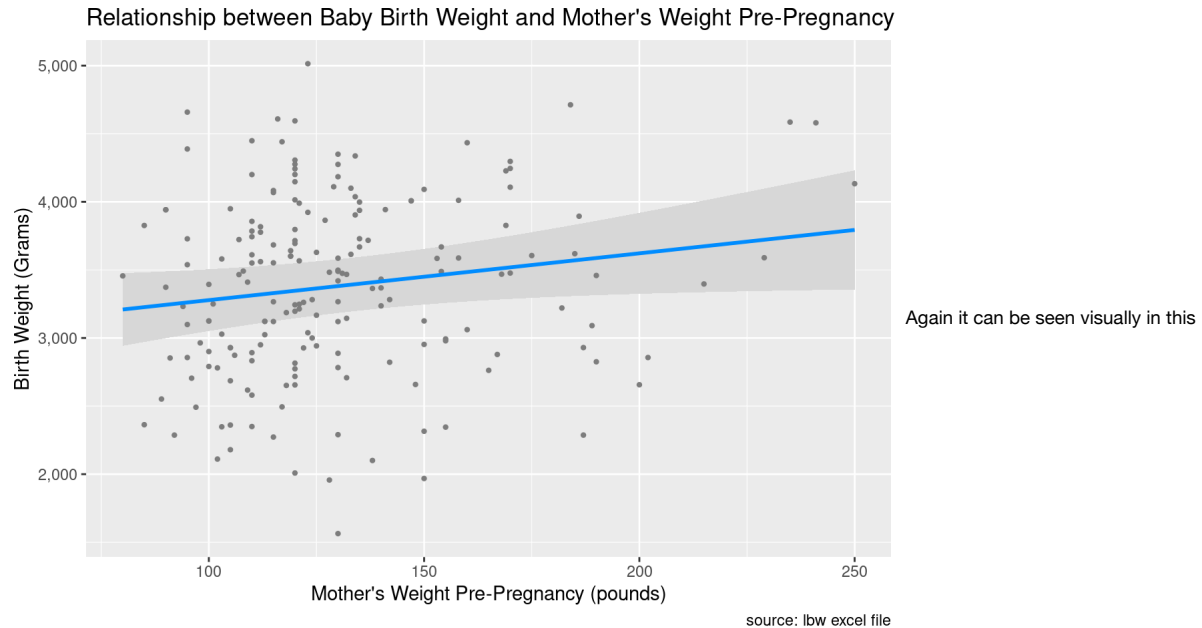
```
##
## Two Sample t-test
##
## data: lwt and bwt
## t = -53.028, df = 376, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -2918.826 -2710.105
## sample estimates:
## mean of x mean of y
## 129.8201 2944.2857
```

```
##
## Pearson's product-moment correlation
##
## data: bwt and lwt
## t = 2.5857, df = 187, p-value = 0.01048
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.04423808 0.32003853
## sample estimates:
## cor
## 0.1857952
```

```
## [1] 0.1857952
```

While the T tests give an extremely low p-value, the pearson correlation gives a much higher p-value. The pearson correlation also calculates a correlation coefficient of approximately 0.1858, which is much more similar to what we see in the graph previously. It would seem that these two variables have very little to do with each other. To be sure let's do a linear regression modelling of the two variables.

Regression Model of lwt and Bwt



graph that although a trendline can be generated, it does look extremely forced. This along with the calculated statistics before, give us evidence which generally indicates no statistically significant correlation between the mother's weight at last menstruation and the birth weight of the baby. This is to say there is not enough evidence to reject the null hypothesis, and for now it has more evidence than the alternative hypothesis.

Data Analysis - Question 6 (Vinton)

Null Hypothesis (H₀): H_0 : There is no statistically significant association between the mother's history of smoking and presence of hypertension in the mother, they occur independent of one another.

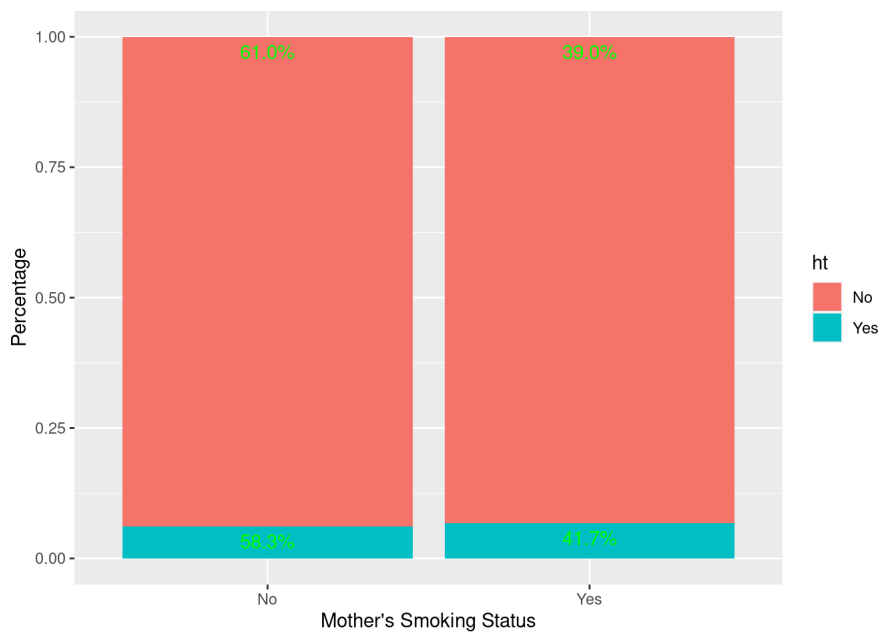
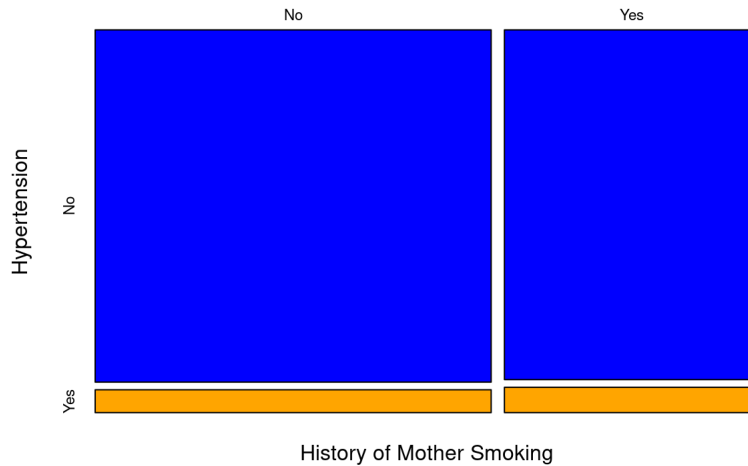
Alternate Hypothesis (H_A): H_A : There is a statistically significant association between the mother's history of smoking and presence of hypertension in the mother, they are related in a statistically significant way.

To investigate whether a mother's history with smoke is statistically related to her having hypertension, let's first create a contingency table to get an idea of the proportions of the matter.

```
##      ht
## smoke No Yes
## No   108 7
## Yes   69 5
```

From this contingency table we can see that there are roughly 1.5 times more mothers who do not smoke in our dataset than those who do. When looking at the mother's who have hypertension, there are also approximately 1.5 times as many who have hypertension in the non-smoking group of mothers than the smoking group. From just a precursory look at the relative proportions, it would seem as though mothers' smoking does not make them any more or less likely to have hypertension, but let's further analyze this with a few statistical tests to be sure. First though, let's get a better visual of the relative proportions that we just found.

Presence of hypertension among smoking & non-smoking mothers



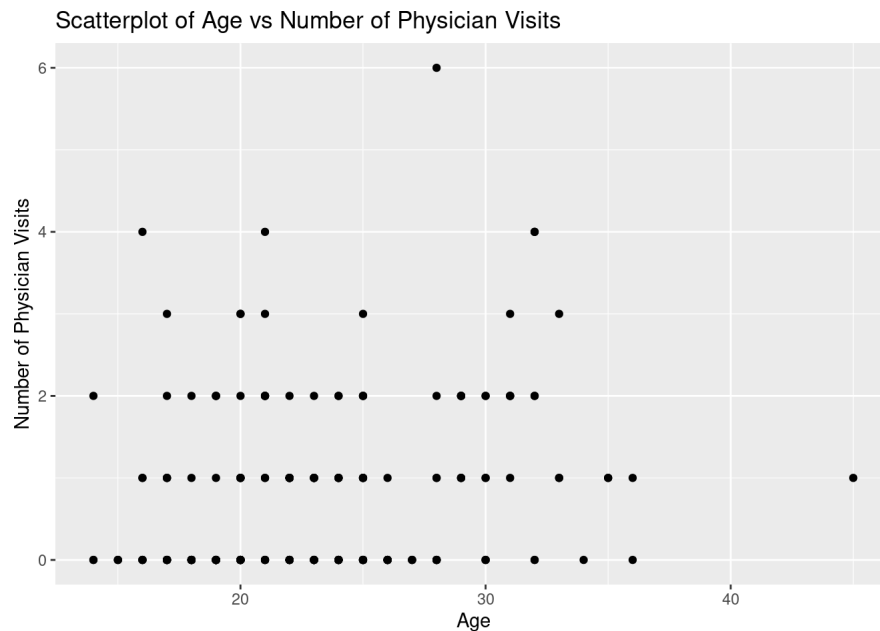
```
##
## Pearson's product-moment correlation
##
## data: smoke_numeric_category and ht_numeric_category
## t = 0.18335, df = 187, p-value = 0.8547
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.1295712 0.1558391
## sample estimates:
## cor
## 0.01340704
```

```
## [1] 0.01340704
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: contingency_table
## X-squared = 1.6837e-31, df = 1, p-value = 1
```

The data from both the T-tests and chi-square gave extremely high p-values, and the chi-square itself was almost zero. In addition the pearson correlation was merely 0.0134. Also these calculated statistics give greater evidence towards the null hypothesis being true. From the data analyzed today, it seems that a mother who has a history of smoking, is not statistically significantly more likely to also have hypertension.

Data Analysis - Question 7 (James)



Data Analysis - Question 8 (James)

- Does the race of a mother (race) affect whether or not a premature entrance into labor occurs (ptl)?

To address this research question, we have created a contingency table between the two categorical variables race (race) and premature labor (ptl) to visualize the frequencies of different combinations of the two variables. We also used a mosaic plot and a bar plot to see the graphical relations between the two variables. After the graphs, we performed a chi-squared test to compare the mean proportions between the two variables. See findings below.

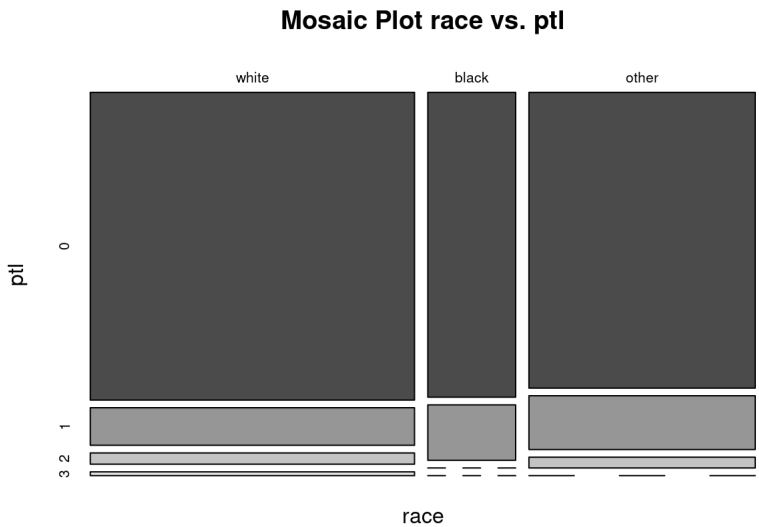
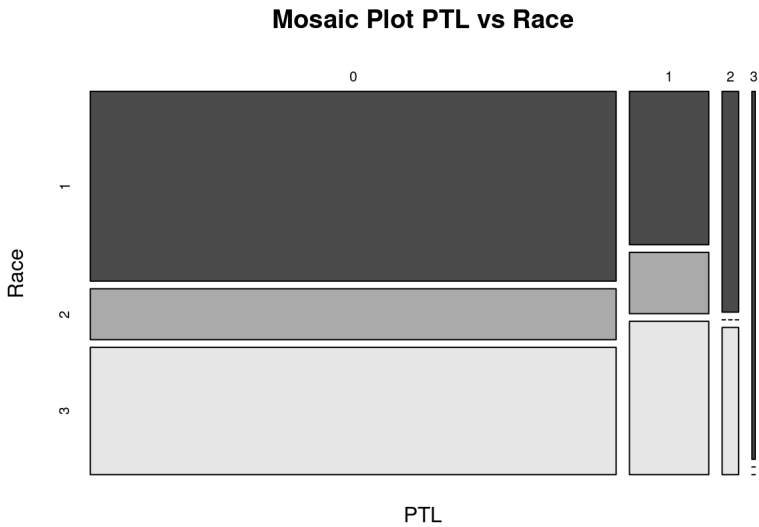
CONTINGENCY TABLE

PTL	Race	Frequencies	Cell.Percentage	Row.Percentage	Column.Percentage
1	0	82	43.3862434	85.416667	85.416667
1	1	22	11.6402116	84.615385	10.416667
1	2	55	29.1005291	82.089552	3.125000
1	3	10	5.2910053	10.416667	1.041667
2	0	4	2.1164021	15.384615	84.615385
2	1	10	5.2910053	14.925373	15.384615
2	2	3	1.5873016	3.125000	0.000000
2	3	0	0.0000000	0.000000	0.000000
3	0	2	1.0582011	2.985075	82.089552
3	1	1	0.5291005	1.041667	14.925373
3	2	0	0.0000000	0.000000	2.985075
3	3	0	0.0000000	0.000000	0.000000

Race	Frequency	Percentage
1	96	50.79365
2	26	13.75661
3	67	35.44974

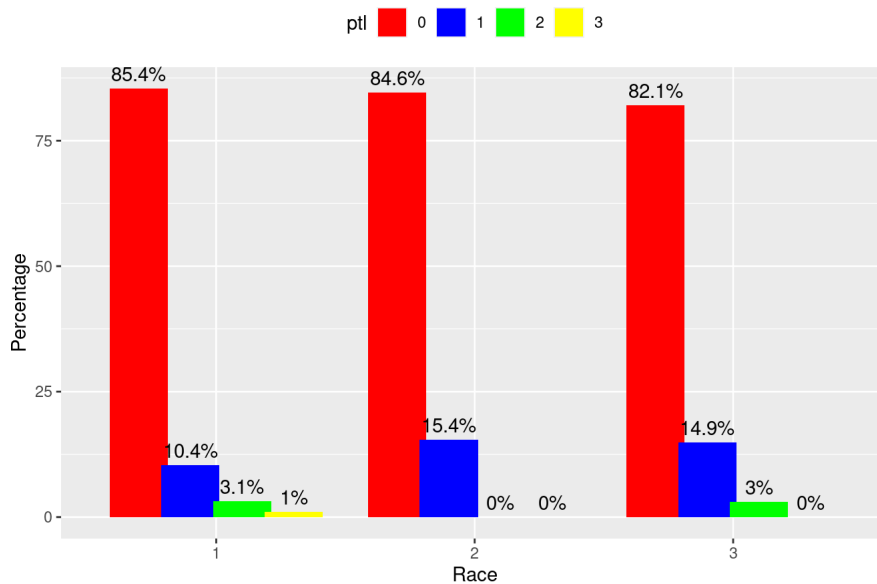
PTL	Frequency	Percentage
0	159	84.1269841
1	24	12.6984127
2	5	2.6455026
3	1	0.5291005

MOSAIC PLOT



BARPLOT

Barplot by Race and Premature Entrances into Labor (Percentage by Row)



Comparing 2 Proportions

Null Hypothesis (H_0): $p_1 = p_2$

- There is no significant difference in the two mean proportions for the variables race and ptl.

Alternate Hypothesis (H_A): $H_A : p_1 \neq p_2$

- There is a significant difference in the two mean proportions for the variables race and ptl.

Significant level: $\alpha = 0.05$

Data Analysis - Question 9 (Josh)

Does a mother's smoking affect the presence of uterine irritability occurrence?

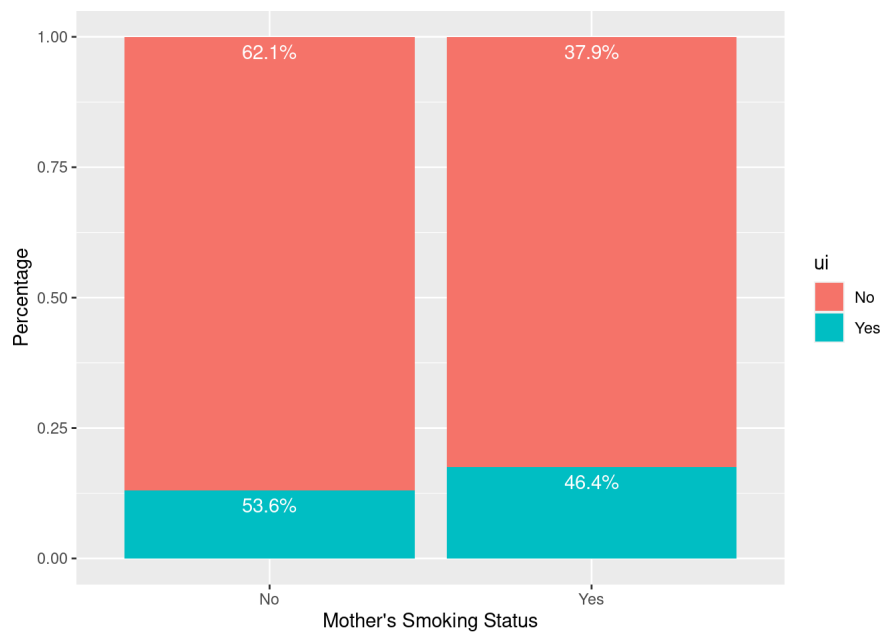
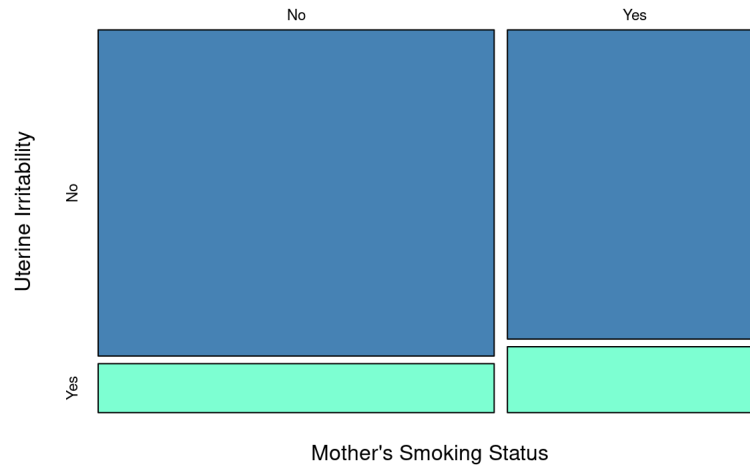
We examine this by comparing the proportionate occurrence of uterine irritability in mothers that smoke versus non-smokers. A chi-squared test is performed to calculate the likelihood that both statistics are independent.

Hypothesis test

Null Hypothesis H_0 : There is no association between smoke and uterine irritability, they occur independent of one another.

Alternative Hypothesis H_A : There is a noticed association between smoke and uterine irritability; they share a relation in their occurrence.

Presence of Uterine Irritability Among Smokers & Non-Smokers



	No Uterine Irritability		Uterine Irritability	Sum
Non-Smoking	100		15	115
Smoking	61		13	74
Sum	161		28	189

The occurrence of uterine irritability in smoking mothers appears to be in higher proportion than in non-smoking mothers. This difference in the sample indicates some weak association between the variables may be present.

A Chi-squared test is performed to validate this observation, with the level of significance drawn at 0.05 or 5%.

The results are a X-squared value of 0.41576 with the p-value equal to 0.5191.

Since the result of the chi-squared test shows a much higher p-value than 0.05, there is not enough evidence to reject the null hypothesis. The present data is insufficient to suggest that a strong dependence between smoke and uterine irritability is present.

Data Analysis - Question 10 (Josh)

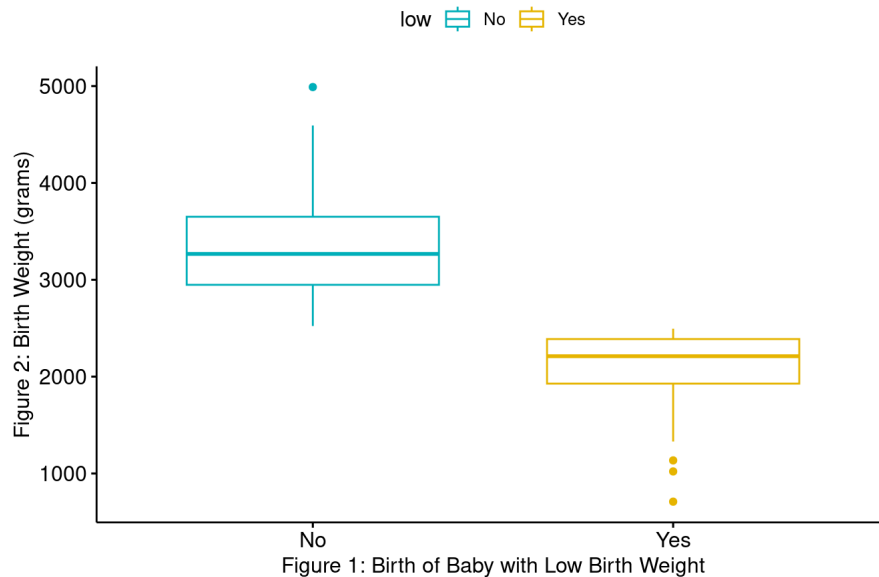
Is there a mean difference between low weight (low) categories according to the birth weight (bwt)?

Significance level, α , will be drawn at 0.05

Null Hypothesis H_0 : There is no difference in the average birthweight from both groups of mothers.

Alternative Hypothesis H_A : There is a noticed difference in the average birthweight from both groups of mothers

Mean Birth-Weight by Low & Non-Low Birth Weight



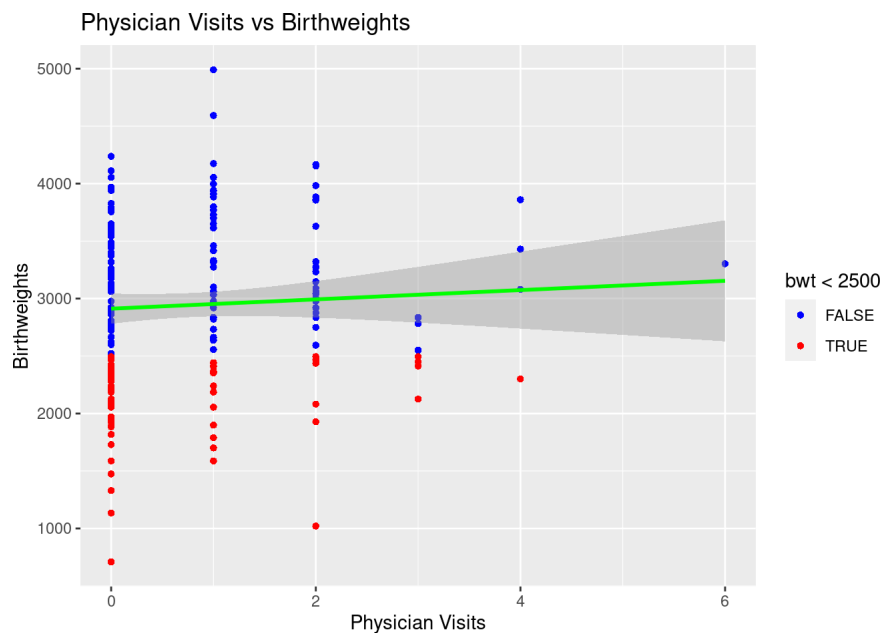
Mean Birth-Weight by Low & Non-Low Birth Weight

Low	Birth Weight Mean
Non-Low	3328.785 grams
Low	2097.085 grams
-----	-----
Difference	1231.7 grams

A t-test is performed to confirm a difference in means, with the calculation resulting in an exceptionally small p-value ($< 2.2e-16$).

The observed difference in mean birthweight between births classified as low-weight and normal weight leads us to reject the null hypothesis.

Data Analysis - Question 11 (Angel)



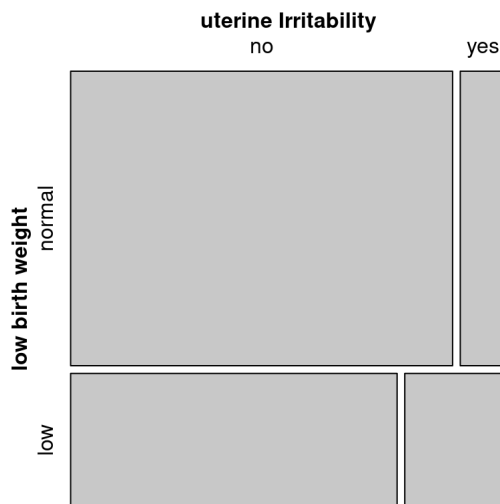
```
##
## Shapiro-Wilk normality test
##
## data: ftv
## W = 0.74625, p-value < 2.2e-16
```

birthweights vs physician visits^^^ (highlighted high and low weights differently)

Data Analysis - Question 12 (Angel)

```
##      ui  No  Yes
## low
## No    116  14
## Yes    45  14
```

Birth Weight vs Uterine Irritability



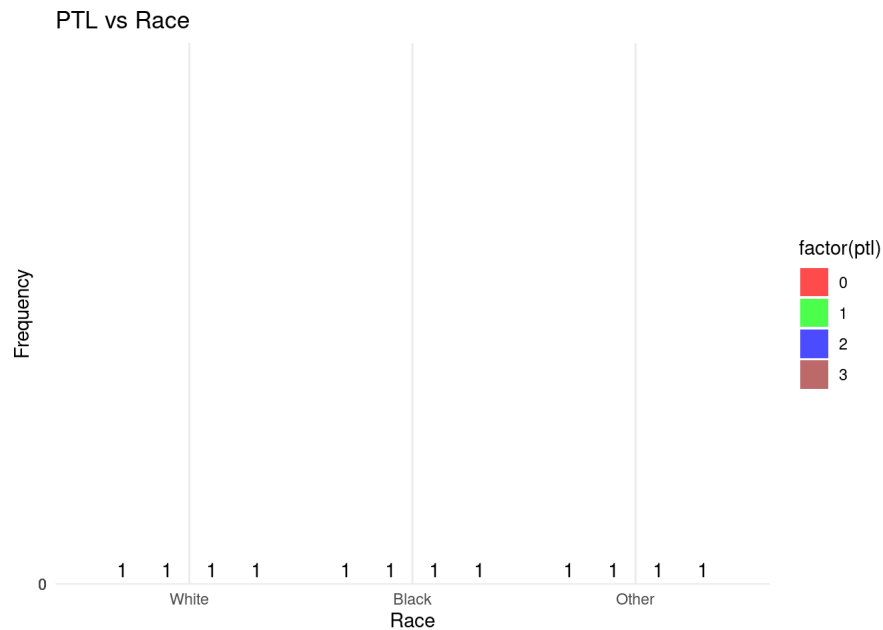
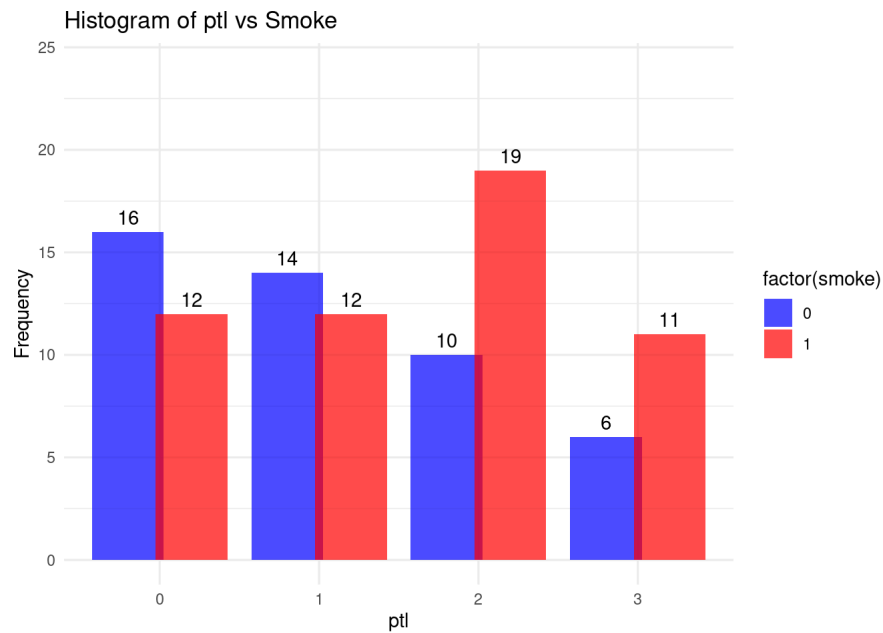
```
##
## Welch Two Sample t-test
##
## data:  bwt by ui
## t = 3.8476, df = 35.669, p-value = 0.000474
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  274.0587 885.4320
## sample estimates:
## mean in group No mean in group Yes
##      3030.174      2450.429
```

```
t_test_result <- t.test(bwt ~ ui, data = lbw)
t_test_result
```

```
##
## Welch Two Sample t-test
##
## data:  bwt by ui
## t = 3.8476, df = 35.669, p-value = 0.000474
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  274.0587 885.4320
## sample estimates:
## mean in group No mean in group Yes
##      3030.174      2450.429
```

birthweights vs ui

Data Analysis - Questions 13 & 14 (Matt did them together)



Results and Discussion (Dori)

After conducting tests and graphical analysis between the various relationships between potential risk factors associated with low birth weight, our results demonstrate that we have found a few variables that are associated as risk factors for delivering low birth weight babies.

1. Smoking is a risk factor associated with low birth weight
2. Race has some relationship with the last weight of mothers before pregnancy, but the last weight of mothers before pregnancy does not affect the birth weight of the baby.
3. Hypertension is a risk factor to low birth weight.
4. Hypertension and smoking does not have much correlation with each other.
5. Mothers who are younger tend to have more physician visits in the first trimester of pregnancy.
6. Mothers who were black were more likely to have 1-3 physician visits compared to mothers who were black.
7. There is no significant data that smoking increases the chances of having uterine irritability.
8. Babies considered low would be weighing under 2500 grams and babies over 2500 grams are considered normal.
9. No specific correlation between physician visits and birth weight.
10. Uterine Irritability does seem to have an affect of birth weight, as the median of mothers who have uterine irritability is significantly lower.
11. The number of false premature labors does not have a clear relationship with race.
12. No clear relationship between false premature labor and mothers who smoke or not.

This is a study that analyzes the socio-demographic factors, maternal factors, and health factors during recent pregnancy that may be associated with low birth weight during delivery.

The maternal age tends to be considered as a key factor for the healthy outcome of pregnancy, as seen with doctors suggesting to deliver babies before the age of 35. Surprisingly, this study revealed that there is no statistical association between the maternal age of mothers and low birth weight of babies, contradicting with the study done in Nepal that shows a higher risk of delivering low birth weight babies by mother age less than 20 and more than 30 years [16, 17, 18]. Smoking during pregnancy does seem to have statistical significance as cigarettes contains harmful chemicals like nicotine that could lead to damage during the development of the baby, leading to cases like low birth weight (Mean birth weight of babies decrease when the mother smokes as seen in the bar plot in Question 1, p-value in t-test also very small rejecting the null hypothesis indicating a significant difference statistically). This finding is consistent with the findings of similar studies done in Bangladesh and Turkey [23, 24].

This study also identified health factors like having hypertension as a potential risk factor for delivering a low birth weight baby. The chances of delivering a low birth baby increases from 29.4% to 58.3%, basically having the chances double. This result is consistent with the results found in a Mendelian Randomization study, grouping hypertension as a potential risk factor for low birth weight babies. Our statistical data for testing the relationship of hypertension and low birth weight reveals the same conclusions, making it seem like a potential risk factor. This might be caused by our sample size.

The races of the women had some significance in determining the last weight of the mothers before pregnancy. Women who were black tend to be heavier than women of white and other races. Mothers with a heavier last weight did not necessarily tend to deliver a singleton baby with a heavier birth weight. This contradicts the study done by the National Library of medicine, categorizing last weight as a factor associated with low birth weight, but last weight is also impacted by socio-demographic factors such as race.

Women who had at least one health problem (hypertension, uterine irritability, etc) during their pregnancy were at a slightly higher risk of delivering low birth weight in comparison to women without any health problem (AOR 2.6, CI: 1.4–4.8). Hypertension and uterine irritability are all potential health risk factors that can result in delivering low birth weight babies (percentages increases). This finding is consistent with the study done in Nepal [17, 20]. In this study, the data has retrieved maternal information namely last weight, physician visits, maternal age, number of premature labors, and birth weight of a baby. Face to face interview was done with the participants for the collection of data using a semi-structured questionnaire, with ethical consent. The total sample size in the study was 369, with 123 cases and 246 controls. The selection of case and controls were random with a random number generator of the exposure status ratio of 1:2 to avoid a biased study. However, the study only involved mothers giving birth inside the Dang District, mothers outside were excluded. The findings could not be generalized as the study was confined in the health institutions of one district. Specific details vary among the population as this study only applied to mothers in the hospital of the Dang district, making it difficult to represent the whole population.

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